Project Description — Science Agent Coworker Hajimi

Global AI4Science Hackathon 2026 · Track: AI Agent · five pages, one argument per page.

1. The problem: papers are read, knowledge is not kept

A researcher entering a new subfield reads twenty to fifty papers. Six months later, three things are gone.

1. **Provenance.** The conclusion survives in memory; the page it came from does not. "Something in that paper improved accuracy by eight points" cannot be checked in time for a rebuttal.
2. **Method.** The method was described in Section 3.2 and then re-implemented by three groups, independently, each with its own bugs.
3. **Judgement.** Which claims are load-bearing and which are incidental lives in one person's head — and leaves when that person does.

Existing tools attack retrieval (search over a corpus) or summarisation (a longer answer). Both are read-only in an architectural sense: ask, receive, forget. Neither leaves behind an **asset** — an artefact that persists, carries its own provenance, and is consumed by the next task instead of being re-derived from scratch.

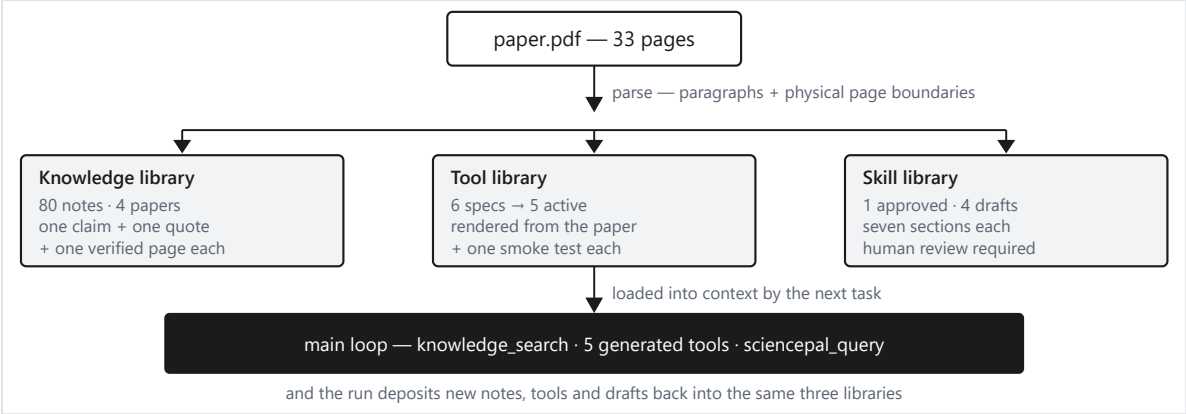
That gap is what this project builds into.

> What if reading a paper left the *repository* richer, not just the conversation?

Who it is for. Graduate students and early-career researchers in computational fields; PIs and systematic-review teams who have to defend where a claim came from. **The scenario we designed against:** twenty papers, one group meeting on Thursday, and a summary that someone will question.

	Chat assistant (today)	This system
What it leaves behind	an answer, in a conversation	notes, tools and skills on disk
Provenance	"the paper says so"	verbatim quote + verified page
Reuse	re-ask, re-read	next task loads them into context
Auditability	none	every artefact reviewable and diffable

2. What we built: a loop that leaves residue



The pipeline. Parsing turns one PDF into three libraries; the main loop consumes them, and a run deposits new artefacts back into the same three.

The loop. A DeepSeek-backed agent loop with an explicit tool registry: `knowledge_search` (query the notes), the five generated method tools, `sciencepal_query` (deep reading of paper text), plus file and shell primitives. Every run is traced into `logs/runs/<id>/` for after-the-fact inspection.

Where SciencePal sits. Not as a wrapper and not as a chat window: one tool node inside our loop, called over three endpoints (`agent/initiate`, `agent-run/{id}/stream`, `agent-run/{id}`). It performs the close reading and returns claims carrying verbatim quotes; our code verifies each quote against the PDF's own text layer and decides what may enter the knowledge base. Be precise about the boundary: the tool path takes paper *text* (up to 8,000 characters per call), while full-PDF runs go through the same three endpoints from a script that uploads the file — both yield the identical claim-and-quote structure. Output is consumed programmatically, never pasted into a document.

What the libraries hold. Atomic notes (one claim, one verbatim quote, one verified page, one confidence level); TypeScript tools rendered from a paper's method paragraph; seven-section skills — domain map, procedure, method-to-tool mapping, failure modes, terminology, evidence index, negative constraints.



The interface, after ingesting four papers: 80 notes, 5 active tools, 5 skills. Each note card carries its claim, its verbatim quote and the page that quote was located on.

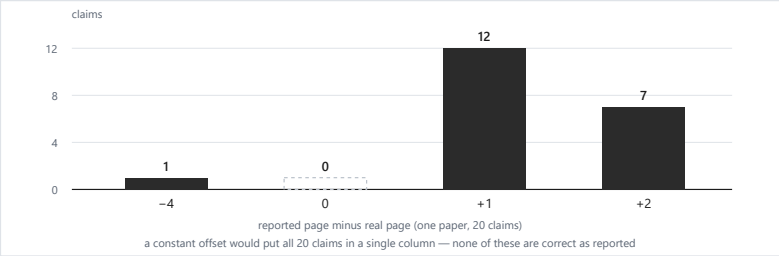
The point. The agent is not the product; the libraries are — the deposit is what compounds.

3. Three decisions that are hard to copy — where the innovation sits

3.1 Provenance is constructed, not reported

We asked a capable reading agent for page numbers. On one paper (ReAct, 20 claims) **all 20 were wrong**, and not by a constant offset (+1 in 12 cases, +2 in 7, −4 in 1) — so no arithmetic correction exists. The quotes, by contrast, were locatable 20 of 20 times.

A second paper reproduced it: Transformer (18 claims), 1 page number right, +1 offset in 16 of 18. Pooled: **38/38 quotes locatable, 1/38 page numbers correct.**



Reported page minus real page, ReAct (20 claims): a constant offset would fill a single column.

Two readings, one entire class of design removed: *no field that the model reports about itself is taken on trust.* Pages are matched from the quote against the PDF's own page boundaries, with a three-level degradation (full sentence → 30 characters → 16) because PDF extraction splits math spans into loose tokens. **99.7%–100%** locate rate over 80 notes.

This is not a feature that can be added late: it fixes the data model (the quote is the anchor, the page is derived), the failure policy (an unlocatable note is stored at `low` confidence and logged), and the interface (every note shows its quote beside its page).

3.2 Tools are grown, not hand-written

A general coding agent ships forty-odd hand-written tools. We ship six that nobody wrote. `toolsynth` extracts a `MethodSpec` from a paper's method paragraph — carrying the evidence page — validates it against a schema, renders TypeScript from a template and generates its smoke test. When no verified implementation exists in the L1 catalogue, the tool degrades to a parameter template for a human to complete, and says so. It never emits plausible-looking code.

3.3 Governance is a runtime gate, not a convention

Four rules are enforced in code rather than documented: a page must be anchored; a claimed failure mode needs at least three samples; a parse quality below 0.7 is quarantined instead of ingested; and a skill cannot become `active` without a named human reviewer. Drafts sit in a separate queue, because a skill that silently becomes active is a skill nobody checked.

3.4 What this costs us

Two costs, stated rather than hidden. **Ingestion cannot invent.** Anchoring needs a text layer: a scanned page, or a paraphrase instead of a quote, produces no note at all — we lose 0.3% of claims rather than mint a wrong citation. **Ingestion spends no tokens.** Extraction is deterministic rules, not a model call per note, so the ingest runs offline at no cost; the trade is that a note's description is its source sentence, not a rewrite.

4. What we verified — and what verifying it caught

Claim	Measurement	How to reproduce
Test suite	**379 passing, 0 failing**, 95 suites, 11.0 s	`npm run verify`
Knowledge library	**4 papers → 80 notes** (77 high / 3 medium)	`node src/main.ts ingest fixtures/papers`
Anchoring	**99.7%–100%** of notes located to a real page	same command, per-paper rate printed
Tool library	6 specs → **5 active** + 1 parameter template	`node src/main.ts tools doctor`
Skill library	**1 approved + 4 drafts**	`node src/main.ts skills list`
Dependencies	**2 runtime** (`zod` , `pdfjs-dist`), 0 agent frameworks	`package.json`

Numbers are cheap; the more useful claim is that the loop found real defects. Three were closed:

1. **Self-reported pages, 1 of 38 correct.** Closed by the anchoring module (§3.1) — and by accepting that a quote, not a page number, is the only thing a model reliably produces.
2. **Half sentences from streaming.** SciencePal emits incremental `chunk` events alongside its final message. Taking the last assistant message blindly yields a truncated quote, which then anchors to nothing and silently lowers yield. We now accept only terminal (`stream_status ≠ chunk`) text.
3. **Arithmetic that cannot be wrong, and was.** A generated `success_rate(11, 10)` returned "110%" — valid syntax, impossible semantics, and it passed its smoke test because the test used only legal inputs. We added cross-field validation plus a regression test (`test/toolsynth/validation.test.ts`).

The third generalises: a tool that *runs* but computes nonsense is worse than no tool, because it looks like a capability.

Inspectable without a key. The 80 notes are frozen in `fixtures/knowledge/` , so both data and interface can be inspected with no API key, no network and no ingestion run:

```
KNOWLEDGE_DIR=fixtures/knowledge node src/main.ts web
```

Running the loop is a different matter, and only partly so: ingesting a PDF still needs no key (§3.4 — rules, not a model call per note), but asking a question needs two credentials the reviewer supplies, since none ship with the repo — `DEEPSEEK_API_KEY` for the agent loop and `SCIENCEPAL_API_KEY` for the SciencePal tool node. All usage is billed to that account.

5. Limitations, and what we would do next

A submission at this stage that claims no limitations would not be credible, so here is the list.

Limitation	Consequence	Next step
Table and figure extraction returns empty arrays	Result-type claims carry their numbers in tables — exactly the claims hardest to anchor; the residual 0.3% failure concentrates here	Convert tables to Markdown before extraction; highest-value single fix
PDF title cleaning	Titles such as `REAC T : SYNERGIZING…` reach the interface	Per-line font-metric normalisation
One method has no verified implementation	`normalize_answer` ships as a parameter template	Grow the verified-implementation catalogue
Page-number finding covers two papers (38 claims)	It settles a design question; it does not estimate an error rate	Repeat across the four-paper set and report the distribution
Web interface runs locally only	Judges cannot click the product without running it	Reviewer package plus the recorded demo; no public deployment, so a stray click cannot consume the API budget
`sciencepal_query` takes text, not a file	One call cannot read a whole PDF; full-paper runs need the script	Move `files[]` upload into the tool, so node and script share one path
No user study	We cannot claim usability findings	Five structured sessions with target users

With another month. Tables first — it unblocks the result half of the knowledge library, the half that a reviewer is most likely to test. Then file uploads into `sciencepal_query`, so the loop can hand over a whole PDF instead of a text window. Then cross-paper consensus: the schema already carries an evidence count and the rule "a single-paper method must be labelled as such", but no skill in the current set has three independent papers behind it, so the mechanism is implemented and unexercised. Then a review workbench, so that the draft queue is something a person can work through rather than a folder of markdown files.

What we would not change. The refusal to trust model-reported structure. Every defect we found this week was a variant of that one, and the whole architecture follows from it.